The Math Behind the Magic
Understanding Modern AI

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2026 AI Connect Conference
March 20, 2026

Overview

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- Why AI Feels Like Magic
- How Language Becomes Numbers
- Vector Embeddings
- Positional Encoding
- Transformers Defined by Self Attention
- Training a Large Language Model
- Generated Language
- Key Takeaways

Introduction

BOEING

GW School of Engineering & Applied Science
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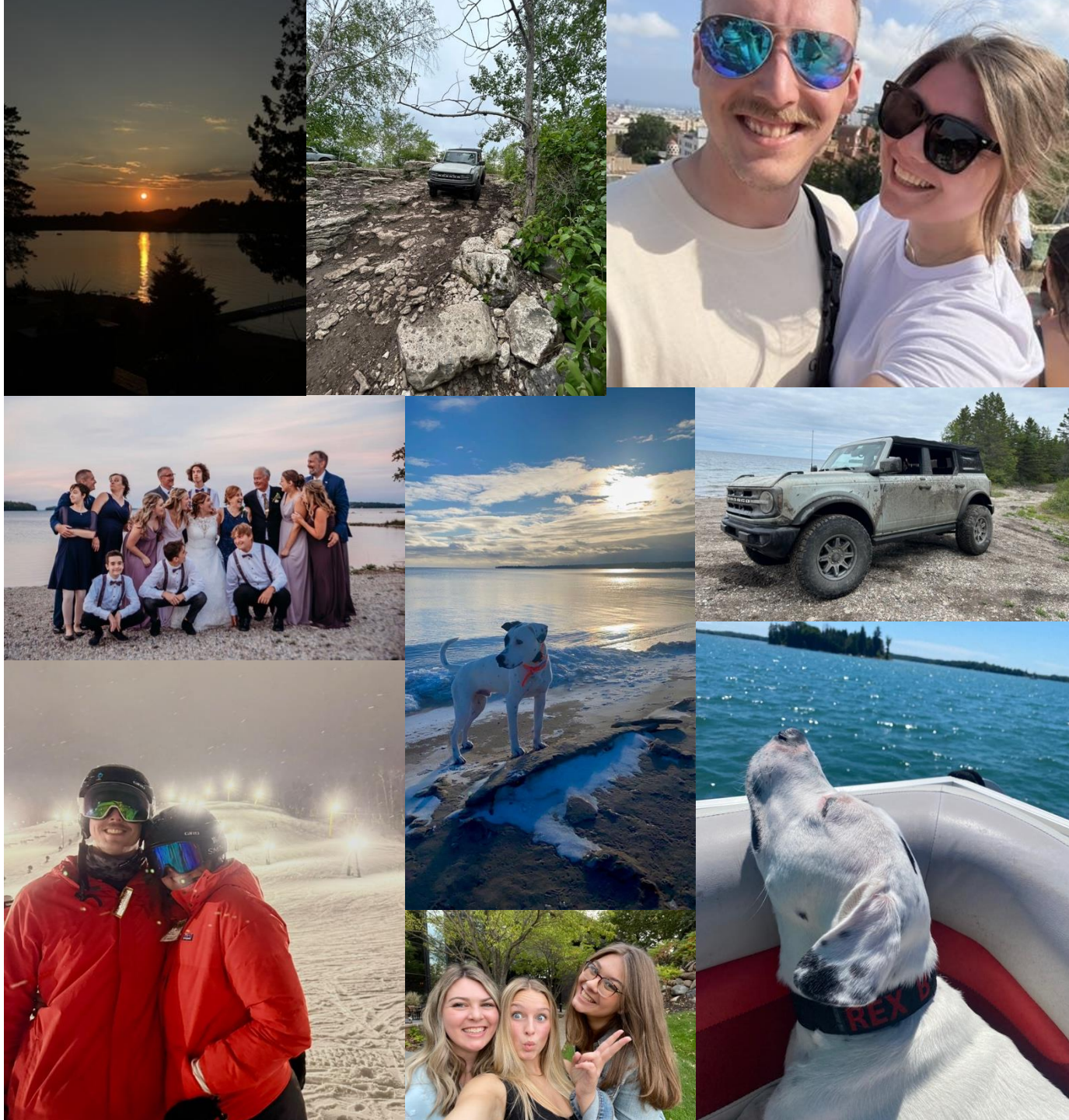
swe

FIRST



Why AI Feels Like Magic

Gemini

Supercharge your creativity and productivity

Chat to start writing, planning, learning and more with Google AI

GPT-5 is here

Our smartest, fastest, and most useful model yet, with thinking built in. Available to everyone.

Empower your organization with Copilot

Powerful generative AI. Enterprise-grade security and privacy. Trusted by companies around the world. What will you achieve with an AI assistant?

What is Perplexity?

Perplexity is your AI-powered answer engine for fast, trustworthy research. It combines live web search with multiple leading AI models to give you up-to-date answers, backed by citations you can verify. Perplexity helps you explore new topics, summarize content, or go deep on complex questions with confidence in where the information comes from.

Think fast, build faster

Brainstorm in Claude, build in Cowork

Chat with Grok — the truth-seeking AI chatbot

Get intelligent answers from Grok. Chat via text or voice with an AI chatbot featuring frontier capabilities in conversation, coding, reasoning, and image and video generation.

1. Humans naturally anthropomorphize machines (Placani, 2024)
2. Natural language triggers social instincts (Cheng et al., 2026)
3. Human like design increases perceived intelligence (Schimmelpfennig et al., 2026)
4. Internal processes are invisible (Mykhaylychenko et al., 2025)
5. Outputs resemble human creativity and reasoning (Cheng et al., 2026)
6. Media framing exaggerates capabilities

It’s not magic… It’s math

How Language Becomes Numbers

Computers cannot understand text, they only understand numbers!

Text

A plane is flying in the sky, a car is driving on the road, and a boat is sailing on the sea.

Tokens

['A', 'Ġplane', 'Ġis', 'Ġflying', 'Ġin', 'Ġthe', 'Ġsky', ',', 'Ġa', 'Ġcar', 'Ġis', 'Ġdriving', 'Ġon', 'Ġthe', 'Ġroad', ',', 'Ġand', 'Ġa', 'Ġboat', 'Ġis', 'Ġsailing', 'Ġon', 'Ġthe', 'Ġsea', '.']

Ġ is the space between the words

Token IDs

[32, 6614, 318, 7348, 287, 262, 6766, 11, 257, 1097, 318, 5059, 319, 262, 2975, 11, 290, 257, 8848, 318, 28499, 319, 262, 5417, 13]

These integers are what the model actually processes

**Tokens generated using GPT 2*

Vector Embeddings

Embeddings represent semantic meaning

Text

A plane is flying in the sky, a car is driving on the road, and a boat is sailing on the sea.

Tokens

['A', 'Ġplane', 'Ġis', 'Ġflying', 'Ġin', 'Ġthe', 'Ġsky', ',', 'Ġa', 'Ġcar', 'Ġis', 'Ġdriving', 'Ġon', 'Ġthe', 'Ġroad', ',', 'Ġand', 'Ġa', 'Ġboat', 'Ġis', 'Ġsailing', 'Ġon', 'Ġthe', 'Ġsea', '.']

Numbers

[32, 6614, 318, 7348, 287, 262, 6766, 11, 257, 1097, 318, 5059, 319, 262, 2975, 11, 290, 257, 8848, 318, 28499, 319, 262, 5417, 13]

Embeddings

[tensor([[-1.4215e-01, -9.0251e-02, -2.0473e-01, -1.1944e-01, -6.5016e-03, -1.7581e-01, -6.8233e-02, 4.7865e-02, -1.2501e-01, -3.2029e-03, 3.9741e-01, -2.1245e-02, -8.7809e-02, 3.7199e-03, -3.9212e-02, -1.6165e-02, -5.1532e-02, -7.7565e-02, 4.2388e-02, -2.2280e-01, ...,]

**Tokens and embeddings generated using GPT 2*

Positional Encoding

Embeddings aren’t enough, the order matters

The **car** followed the **truck**
The **truck** followed the **car**

Positional Encoding

- Each pair (sin(θ),cos(θ)) lies on the unit circle
- As position increases, the point rotates around the circle
- Distance between tokens = angle distance
- Each dimension pair = rotating vector

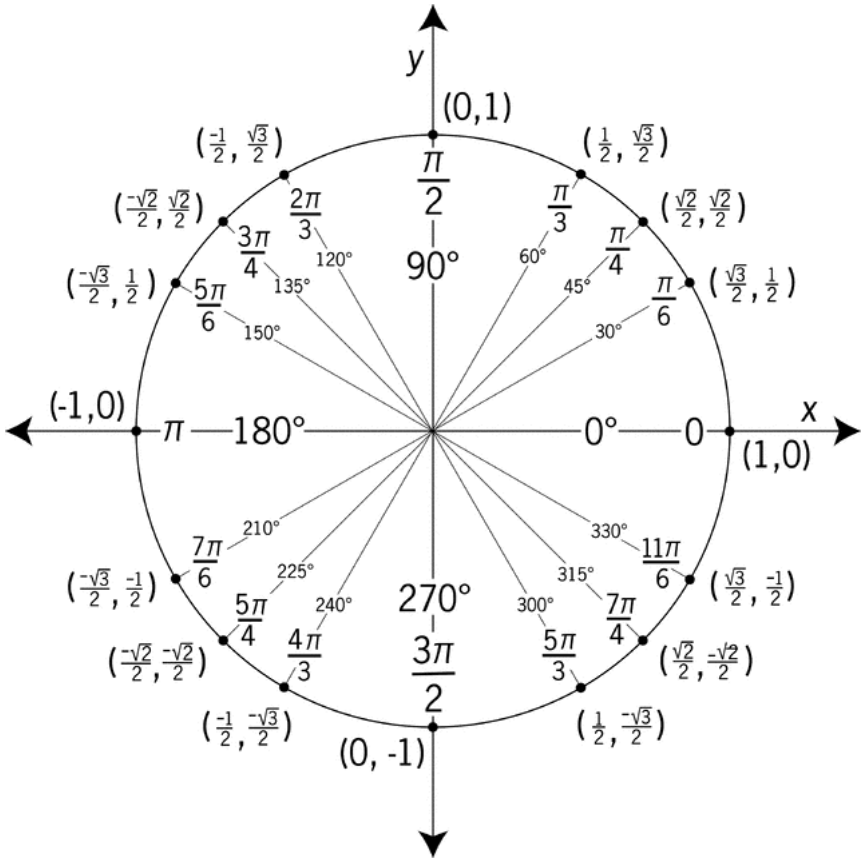
$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \qquad PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d}}}\right)$$

Why this works

$$sin(a+b) = sin(a)cos(b) + cos(a)sin(b)$$

$$cos(a+b) = cos(a)cos(b) - sin(a)sin(b)$$

- Using this trig identity a shift in position becomes a linear transformation
- Their encodings differ by a known rotation



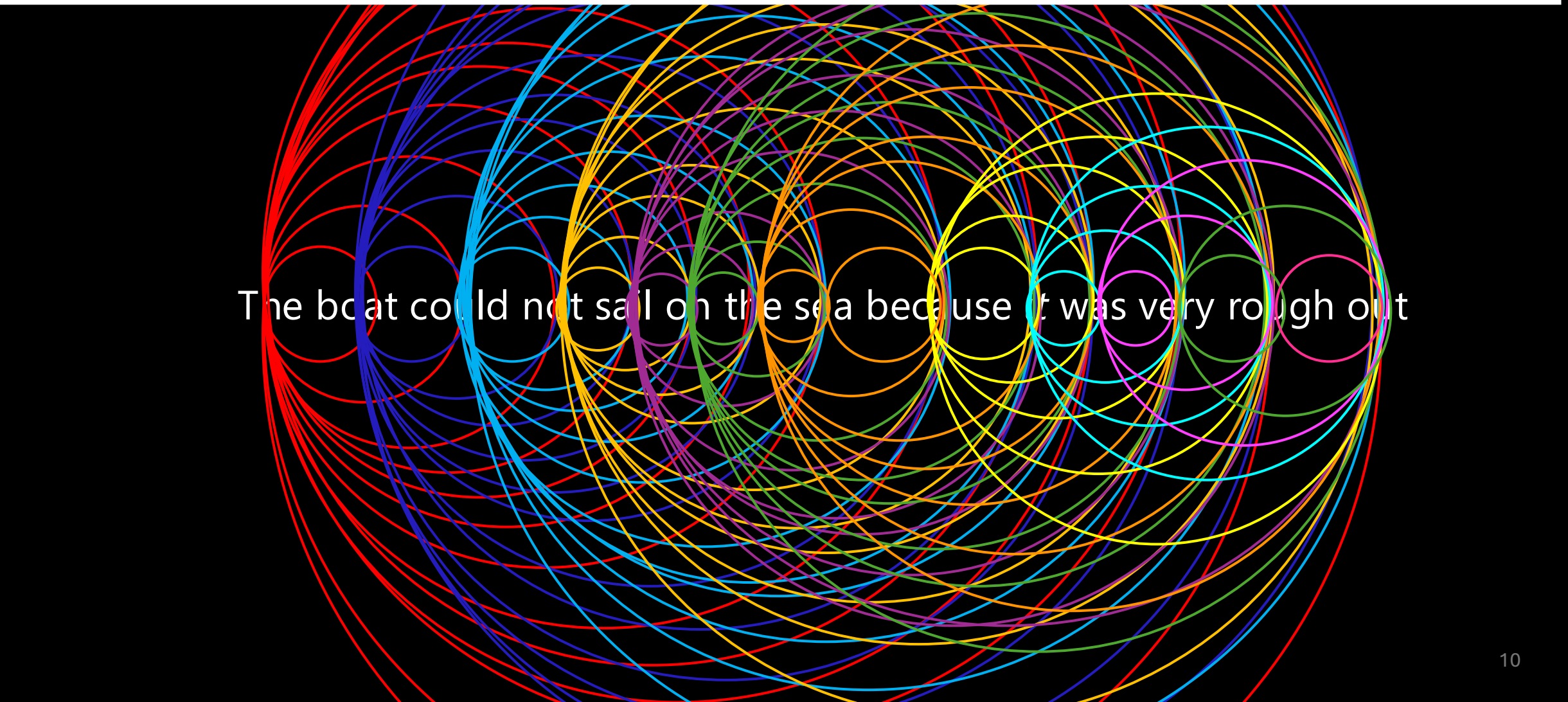
Transformers Defined by Self Attention

"The boat could not sail on the sea because *it* was very rough out"

Transformers use self-attention to automatically determine "it"

Transformers Defined by Self Attention

Each element pays attention to every other element



Transformers Defined by Self Attention

$$\boldsymbol{Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V}$$

where $\sqrt{d_k}$ is a scaling factor, usually based on the individual head dimension or number of heads.

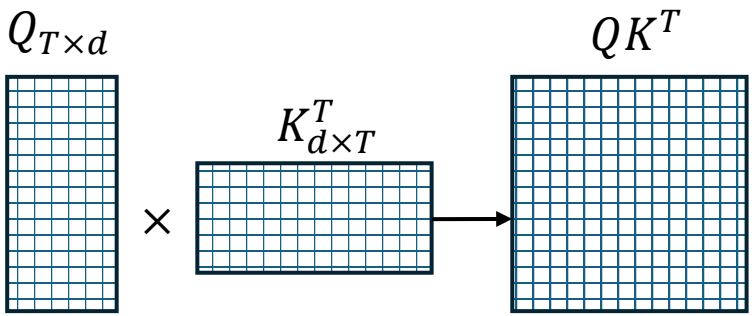
Convert Embeddings into Q, K V vectors → **Compute Similarity Scores** → **Normalize with Softmax** → **Use weights to combine information**

Q = Query → What the token looks for
K = Key → What the token contains
V = Value → Information itself

Compute the dot products between queries and keys

Convert the scores to probabilities

Using the weighted information from other tokens update the representation



Training a Large Language Model (LLM)

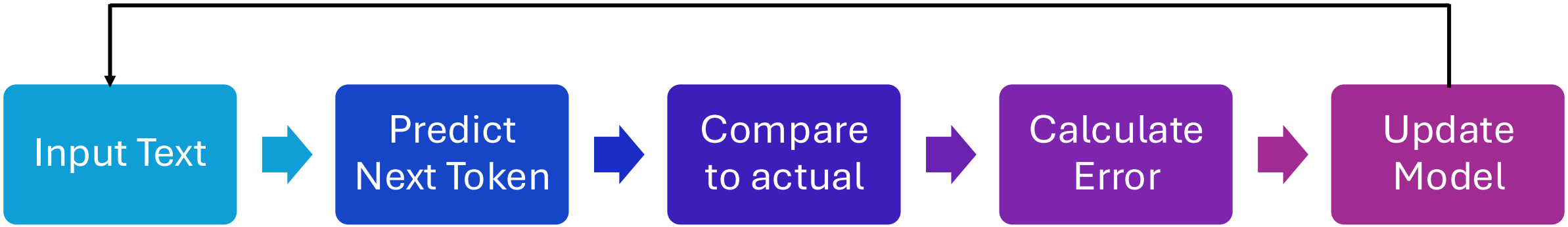
A plane is flying in the __Sky__

A car is driving on the __Road__

A boat is sailing on the ~~Sea~~ **River**

A boat is sailing on the __River__

Training a Large Language Model (LLM)



A boat is

running

running | **sailing**

Difference between "running" and "sailing" vector spaces

Adjust weights slightly (backpropagation)

Generating Language

The inference pattern that we all know and love

Tokenize Prompt → **Run Transformer** → **Compute next token probabilities** → **Select next token** → **Add to response**

User passes a prompt to the model

Transformer calculates self attention

Probabilities of next possible tokens

Token with highest probability is selected

Token is added to the response

Key Takeaway

AI is not magic… it is mathematics!

Text → Tokens → Vectors → Transformers → Training → Prediction

Why it is important?

1. It's important to understand the tools that you use everyday
2. Helps reduce the perceived intelligence of the model

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention is all you need*. arXiv. https://arxiv.org/abs/1706.03762

Cheng, M., Lee, A.Y., Rapuano, K. *et al*. Metaphors of AI indicate that people increasingly perceive AI as warm and human-like. *Commun Psychol* 4, 8 (2026). https://doi.org/10.1038/s44271-025-00376-6

Placani, A. Anthropomorphism in AI: hype and fallacy. AI Ethics 4, 691–698 (2024). https://doi.org/10.1007/s43681-024-00419-4

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